

Replication / Machine Learning

[Re] Accuracy is Not All You Need on Current Small Instruction-Tuned Models

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Abstract We reproduce the central finding of Dutta et al. (2024), that quantizing a language model can change a large fraction of its individual answers while leaving benchmark accuracy almost unchanged, and we test how it behaves on a newer and smaller generation of models. Using the original bitsandbytes 8-bit and 4-bit conditions and the original metric definitions, we evaluate five current small instruction-tuned models (Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-3B, Gemma-2-2B, and Phi-3.5-mini) on ARC-Challenge, MMLU, and CommonsenseQA. The core result reproduces: flips are substantial while accuracy is roughly preserved, and lower-bit quantization produces more flips. On these models, however, two of the original quantitative claims are weaker. Accuracy is preserved less well, with 4-bit drops of up to five points on the smallest model, and the balance between correct-to-incorrect and incorrect-to-correct flips shifts toward loss, to about three to two pooled across models. This shift remains below the three to one that a four-option task would produce by chance, and the elevated rate of recovered answers the original describes therefore persists, though it is weaker here and accounts for the larger accuracy loss. Using the original AllFlips metric, we confirm that wrong-to-wrong transitions are a large share of behavioral change, and a determinism control shows the measured flips are not run-to-run noise. The balance between gaining and losing correct answers has shifted toward loss on this generation, and accuracy alone does not reveal it.

A replication of Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. Accuracy is Not All You Need. Advances in Neural Information Processing Systems (NeurIPS), 2024.

1 Introduction

Dutta et al.^[1] observed that quantizing a language model can leave its benchmark accuracy almost unchanged while altering a large fraction of its individual answers. They compared each quantized model against its full-precision baseline, counted the answers that changed between correct and incorrect, and called these changes flips. Because flips can reach double digits while accuracy moves by a point or less, accuracy alone does not reveal whether a quantized model behaves like the one it replaces. The finding is now widely cited and often treated as an established property of quantization.

The original study used 2023-era models, the Llama2-chat (7B, 13B, 70B) and Yi-chat (6B, 34B) families. These models are larger than, and were trained differently from, the small distilled models that practitioners quantize and run on consumer hardware today, which sit at or below three billion parameters. Whether the flips phenomenon behaves the same way on this newer generation had not been examined.

We replicate the bitsandbytes portion of Dutta et al.^[1] on five current small instruction-tuned models across three multiple-choice benchmarks, and the core phenomenon holds. Quantization changes a substantial fraction of answers while accuracy stays roughly stable, and lower-bit quantization changes more. On this generation, however, two of the

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Code is available at <https://github.com/ab-ch90/quantization-flips-small-llms> – DOI 10.5281/zenodo.21802395.

Data is available at <https://github.com/ab-ch90/quantization-flips-small-llms>.

original quantitative claims are weaker. Accuracy is preserved less well, and the balance between correct-to-incorrect and incorrect-to-correct flips shifts toward loss, although it remains below the level that a four-option task would produce by chance. Wrong-to-wrong transitions, which the original reports separately as AllFlips, account for a large share of behavioral change. The net effect is that the balance between gaining and losing correct answers has moved toward loss, and accuracy alone does not reveal it.

2 Scope and claims under test

We organize the replication around five claims from the original work:

- C1. Under quantization, benchmark accuracy is nearly preserved, to within roughly one to two percent.
- C2. Flips are substantial even when accuracy is preserved.
- C3. Lower-bit quantization produces more flips than higher-bit quantization.
- C4. Correct-to-incorrect and incorrect-to-correct flips are roughly balanced, which is what allows accuracy to remain stable as flips accumulate.
- C5. Flips concentrate on items where the model's probability margin between its top two options is small.

We test the bitsandbytes 8-bit and 4-bit conditions on three of the original benchmark types, using the original metric definitions. We do not cover the GPTQ, AWQ, or SmoothQuant conditions or the generative-task analysis; the scope of this replication is the core quantization-flips result.

3 Methods

3.1 Models

We evaluate five current small instruction-tuned models, each of which fits in 16-bit precision on a single 8 GB consumer GPU, which makes the full-precision baseline computable locally: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct [2], Llama-3.2-3B-Instruct [3], Gemma-2-2B-it [4], and Phi-3.5-mini-instruct [5]. These models span four families and two parameter scales below and around three billion. We chose breadth across families over breadth across sizes, because the claim under test concerns quantization behavior in general.

3.2 Quantization

Each model is quantized with bitsandbytes [6] at 8-bit and 4-bit, the same library and bit widths that the original study uses for its bitsandbytes conditions. The 4-bit configuration uses the NF4 type with a 16-bit compute dtype, and the 16-bit model serves as the reference for all comparisons.

3.3 Benchmarks

We evaluate on three multiple-choice benchmarks with objective answer keys: ARC-Challenge [7], MMLU [8], and CommonsenseQA [9]. These overlap with the original benchmarks and span science reasoning, broad knowledge, and commonsense reasoning. For MMLU, we use a fixed set of eight subjects chosen before running, two from each of its four categories, to give subject breadth without the full suite (Appendix). We keep only standard four-option items. The fixed subsets contain 1,000 ARC-Challenge items, 975 MMLU items, and 986 CommonsenseQA items.

3.4 Reading the answer

We read each model’s answer from its probability over the four option-letter tokens at the answer position, taking the option with the highest probability. Reading the letter probabilities directly avoids the measurement noise introduced by parsing generated text. Because decoding is greedy, a model produces the same prediction every time it runs on the same item in the same setup.

3.5 Metric definitions

We follow the original metric definitions and extend them. Flips counts the answers that move from correct to incorrect together with those that move from incorrect to correct, and excludes answers that move from one wrong option to another. AllFlips adds those wrong-to-wrong transitions. We report both, together with the directional breakdown, the separate counts of correct-to-incorrect and incorrect-to-correct, because the balance between those two directions is where we find the clearest divergence.

3.6 Setting differences from the original study

Our setup differs from that of Dutta et al.^[1] most of all in the models, which are a newer and more distilled generation. Two further differences bear on how directly the numbers compare. Our evaluation is zero-shot, whereas the original MMLU numbers are five-shot, and we evaluate a fixed subset of each benchmark instead of the full set. For these reasons, we do not claim exact numerical replication of the reported flip rates. We claim instead that the phenomenon and its qualitative claims replicate, and we report where specific claims diverge.

3.7 Determinism control

The flips metric compares a quantized run against a single full-precision baseline run and treats every difference as a consequence of quantization, which is valid only if the baseline is stable. Recent work shows that language model inference can be non-deterministic even at temperature zero, mainly because of batch-dependent numerics in serving [10, 11]. We checked this directly by running the full-precision model on the same items four times with nothing changed. On our setup, the floor is zero. Every item received the same prediction across all four runs, and single-item and batched passes agreed. Varying the batch size across runs changed a single item out of three hundred, a low-margin question that tipped under a tiny numeric change. We therefore treat the baseline as stable and the flips we report as consequences of quantization. Neither the original study nor its successors report this control.

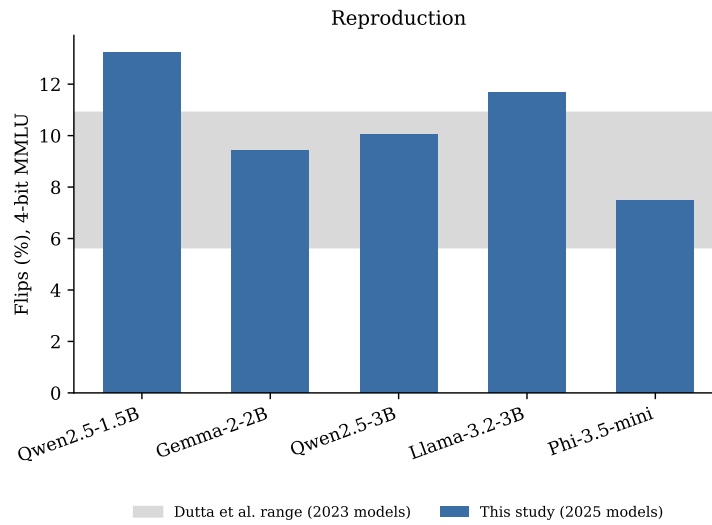


Figure 1. Replication of the flips finding on current small models. Bars show the 4-bit MMLU Flips rate for each model; the shaded band marks the 5.7 to 10.9 percent range Dutta et al. report for their 2023 chat models on the same metric (their Table 5). The smaller models we test sit at the upper end of that range or above it, consistent with the original’s finding that smaller models show more flips.

4 Results

4.1 Claims that replicate

The central finding that flips are substantial while accuracy is preserved (C2) replicates. Across all five models on MMLU, 4-bit quantization produces a mean Flips rate of 10.4 percent while accuracy changes little. The original reports 4-bit MMLU Flips of 5.7 to 10.9 percent for its 2023 chat models (their Table 5), and our small models sit at the upper end of that range or above it, from 7.5 percent for Phi-3.5-mini to 13.2 percent for Qwen2.5-1.5B (Figure 1), consistent with the original’s own finding that smaller models show more flips. Flips are present under conditions where accuracy alone would suggest that the quantized model is an equivalent replacement.

The dose-response claim (C3) holds for every model at the benchmark mean. Averaged over the three benchmarks, 4-bit quantization produces more flips than 8-bit in all five models (Figure 2), and the ordering holds on 14 of the 15 individual model-benchmark conditions. The single exception is Phi-3.5-mini on CommonsenseQA, where 4-bit produces marginally fewer flips than 8-bit (4.7 against 4.9 percent). Lower-bit quantization moves the model further from its full-precision behavior, as the original study found.

The headline result therefore replicates. Accuracy is not sufficient to determine whether a quantized small model behaves like its full-precision baseline, and the gap between preserved accuracy and changed answers is as real on current models as on the previous generation. Full per-model numbers appear in Table 1.

4.2 Claims that weaken on smaller models

Two of the enumerated claims are weaker on these models, and a third property, wrong-to-wrong transitions, is substantial here. The differences are consistent across models,

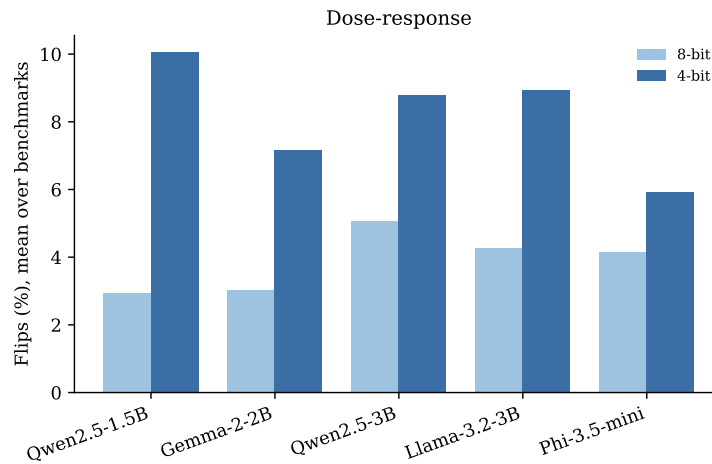


Figure 2. Mean Flips rate over the three benchmarks at 8-bit and 4-bit. Heavier quantization produces more flips for every model.

and all indicate the same conclusion: that quantization is harder on these distilled small models than on the larger models the original tested.

Accuracy is less preserved on smaller models (C1 weakens). The original work found accuracy preserved to within roughly one to two percent. On these models, the loss is larger, with the largest 4-bit accuracy change reaching 5.0 points and several conditions dropping three to five points. The effect is strongest on the smallest model, Qwen2.5-1.5B, which loses 5.0 points on ARC-Challenge and 4.4 points on MMLU under 4-bit quantization. This finding extends the original instead of contradicting it, because its own results already show smaller models losing more accuracy and showing more flips than larger ones (their Table 5 and observation 5 in Section 4.1; for example, Llama2-7B loses 1.5 points on 4-bit MMLU against 0.8 for Llama2-70B), and the sub-3B models we test sit at the far end of that trend. Even 8-bit quantization, which the original found nearly lossless, moves accuracy on these models and already produces several percent flips.

The correct-to-incorrect balance shifts toward loss (C4 weakens). The original study found the two flip directions roughly balanced, which is what allows accuracy to stay stable while flips accumulate. That balance is itself notable, because on a four-option question a correct answer has three ways to become incorrect and an incorrect answer only one way to become correct, and chance alone would therefore push the ratio toward three to one. The original result sits well below that level, which indicates that correct answers are protected by a larger probability margin and that incorrect-to-correct transitions occur more often than the four-option structure would predict.

On these smaller models, the balance shifts toward correct-to-incorrect but does not reach that structural baseline (Figure 3). Pooled across all five models and three benchmarks at 4-bit, quantization produces 738 correct-to-incorrect flips against 470 in the other direction, a ratio near three to two that a two-sided binomial test against an even split rejects decisively ($p \approx 1 \times 10^{-14}$). The strength of the shift varies across models. Pooling each model over its three benchmarks, the shift is clear for Qwen2.5-1.5B (199 against 98, $p < 0.001$), Qwen2.5-3B (167 against 93, $p < 0.001$), and Phi-3.5-mini (107 against 68, $p = 0.004$), and weaker and not individually significant for Llama-3.2-3B (147 against 117, $p = 0.07$) and Gemma-2 (118 against 94, $p = 0.11$). Pooled and at the model

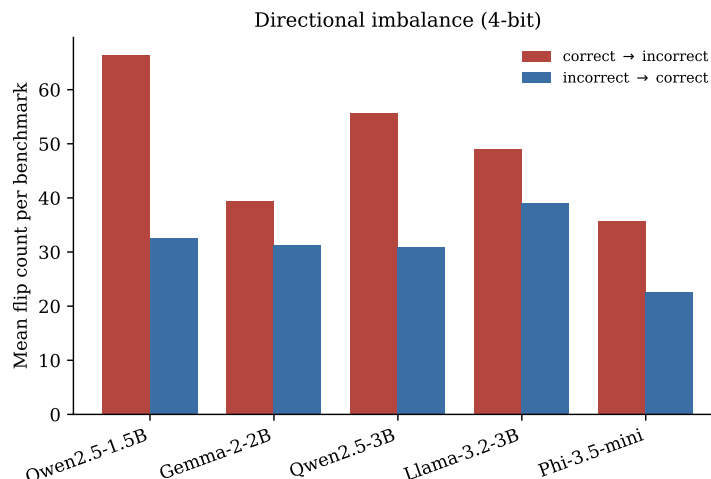


Figure 3. Directional imbalance of flips under 4-bit quantization, averaged over the three benchmarks. Bars show the mean count of correct-to-incorrect and incorrect-to-correct flips per benchmark. Every model leans toward correct-to-incorrect, strongly for the Qwen models and Phi-3.5-mini and weakly for Llama-3.2-3B and Gemma-2. The balance sits between the original study’s near-even result for larger models and the three-to-one a four-option task would produce by chance.

level, the ratio stays below the structural three to one, near three to two overall and at most about two to one for any single model. On 4-bit MMLU, Qwen2.5-1.5B shows 86 correct-to-incorrect flips against 43, Qwen2.5-3B 66 against 32, and Phi-3.5-mini 46 against 27. The one condition that reaches the structural baseline is the smallest model on ARC-Challenge, Qwen2.5-1.5B at 68 against 18, where the protection appears to break down.

The two studies are best read as points on a spectrum. The larger 2023 models sit near balance, well below the structural three to one, while these smaller models sit partway toward it. Incorrect-to-correct transitions remain more common than chance in both cases, and the effect the original describes therefore still holds, though the protection is weaker here, and the residual tilt toward correct-to-incorrect is what drives the accuracy loss noted above.

Wrong-to-wrong transitions are a large share of behavioral change. The original study reports these transitions separately, in its AllFlips metric (their Figure 8 and Table 12), and excludes them from Flips to keep the metric applicable to generative tasks, where a wrong-to-wrong change is not well defined. On these models, the gap between the two metrics is large, with wrong-to-wrong transitions amounting to a third to a half as many as the directional flips. On 4-bit MMLU, Qwen2.5-1.5B has 57 wrong-to-wrong transitions against 129 directional flips, and Gemma-2 has 42 against 92. A user comparing the full-precision and quantized models by AllFlips rather than Flips would see substantially more changed answers, the same relationship that the original AllFlips results show.

4.3 Confidence profile (C5)

The directional shift raises a question the counts alone do not answer. When quantization turns a correct answer wrong, was the full-precision model confident, or was the item a close call that quantization tipped? We examined this on the model with the

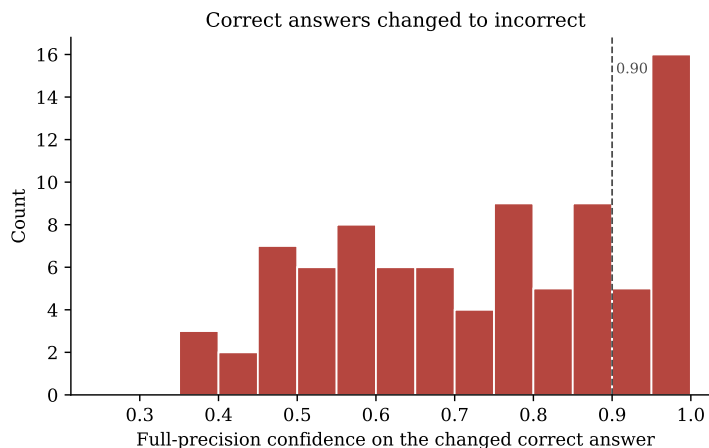


Figure 4. Full-precision confidence on the correct answers that 4-bit quantization changed to incorrect, for Qwen2.5-1.5B on MMLU. Most were weakly held, with a minority above 0.90 confidence that the full-precision model was sure about.

strongest imbalance, Qwen2.5-1.5B on MMLU (Figure 4). Most correct-to-incorrect flips occur on items where the full-precision model was correct but weakly committed, with the probability on its chosen option near the level at which a small perturbation could change the outcome, which matches C5. However, of the 86 correct-to-incorrect flips, 21 occurred on items where the full-precision model placed at least 0.90 probability on the correct option. In those cases the full model was confident and correct, and quantization changed the answer nonetheless. C5 therefore holds for the bulk of flips but not for all of them.

5 Discussion

On this generation, the flips phenomenon replicates and extends to newer models, and the balance between gaining and losing correct answers shifts toward loss. A practitioner who quantizes one of these models and checks only accuracy will not see how much its behavior has changed, and here the net effect on accuracy is worse than the near-even picture the original study reports for larger models.

The accuracy loss and the shifted balance are two views of a single effect. Accuracy is preserved less well because the flips tilt toward correct-to-incorrect. The confidence analysis indicates why this happens and largely replicates C5. Most of the lost correct answers were held weakly, near the margin at which a small perturbation can change the outcome, while a minority involved answers the full model held with high confidence, which the margin mechanism of C5 does not by itself explain. The wrong-to-wrong transitions add further behavioral change on top of the directional flips.

A plausible account ties the two studies together. These distilled small models have less redundancy than the larger 2023 models, and the error that quantization introduces is therefore more likely to change an outcome and less likely to be absorbed. This is the same margin mechanism the original study uses to explain why incorrect-to-correct transitions exceed chance. On smaller models, the margins are thinner, and the protection is therefore weaker, with the balance sitting closer to the structural baseline, which explains both the larger accuracy losses and the shift toward correct-to-incorrect. We offer this as an interpretation consistent with the measurements, not as a demonstrated

Table 1. Full per-model, per-benchmark, per-precision results. Accuracy at full precision (fp16 Acc) and after quantization (Q Acc), the accuracy change (Δ Acc), Flips and AllFlips as percentages, and the directional breakdown of changed answers (C→I correct-to-incorrect, I→C incorrect-to-correct, W→W wrong-to-wrong) as counts. The benchmark subsets contain 1,000 ARC-Challenge, 975 MMLU, and 986 CommonsenseQA items, which keeps the counts comparable across rows.

Model	Benchmark	Prec	fp16 Acc	Q Acc	Δ Acc	Flips	AllFlips	C→I	I→C	W→W
Qwen2.5-1.5B	ARC-Challenge	8-bit	75.3	76.2	+0.9	1.9	2.8	5	14	9
Qwen2.5-1.5B	ARC-Challenge	4-bit	75.3	70.3	-5.0	8.6	11.1	68	18	25
Qwen2.5-1.5B	MMLU	8-bit	61.0	61.2	+0.2	4.3	5.8	20	22	15
Qwen2.5-1.5B	MMLU	4-bit	61.0	56.6	-4.4	13.2	19.1	86	43	57
Qwen2.5-1.5B	CommonsenseQA	8-bit	77.5	77.9	+0.4	2.6	2.9	11	15	3
Qwen2.5-1.5B	CommonsenseQA	4-bit	77.5	76.7	-0.8	8.3	10.1	45	37	18
Gemma-2-2B	ARC-Challenge	8-bit	74.4	74.5	+0.1	2.9	4.1	14	15	12
Gemma-2-2B	ARC-Challenge	4-bit	74.4	73.2	-1.2	6.4	8.7	38	26	23
Gemma-2-2B	MMLU	8-bit	57.0	57.9	+0.9	3.6	5.2	13	22	16
Gemma-2-2B	MMLU	4-bit	57.0	57.0	+0.0	9.4	13.7	46	46	42
Gemma-2-2B	CommonsenseQA	8-bit	74.9	73.9	-1.0	2.6	3.0	18	8	4
Gemma-2-2B	CommonsenseQA	4-bit	74.9	73.7	-1.2	5.7	7.4	34	22	17
Qwen2.5-3B	ARC-Challenge	8-bit	82.8	82.4	-0.4	3.8	5.1	21	17	13
Qwen2.5-3B	ARC-Challenge	4-bit	82.8	80.2	-2.6	7.8	10.0	52	26	22
Qwen2.5-3B	MMLU	8-bit	65.4	64.0	-1.4	5.9	8.2	36	22	22
Qwen2.5-3B	MMLU	4-bit	65.4	61.9	-3.5	10.1	14.9	66	32	47
Qwen2.5-3B	CommonsenseQA	8-bit	81.6	81.6	+0.0	5.5	6.2	27	27	7
Qwen2.5-3B	CommonsenseQA	4-bit	81.6	80.2	-1.4	8.5	9.8	49	35	13
Llama-3.2-3B	ARC-Challenge	8-bit	71.5	71.2	-0.3	2.9	3.9	16	13	10
Llama-3.2-3B	ARC-Challenge	4-bit	71.5	70.2	-1.3	7.9	11.1	46	33	32
Llama-3.2-3B	MMLU	8-bit	59.5	57.8	-1.6	6.4	8.3	39	23	19
Llama-3.2-3B	MMLU	4-bit	59.5	58.3	-1.2	11.7	16.1	63	51	43
Llama-3.2-3B	CommonsenseQA	8-bit	74.8	74.5	-0.3	3.5	4.5	19	16	9
Llama-3.2-3B	CommonsenseQA	4-bit	74.8	74.3	-0.5	7.2	8.7	38	33	15
Phi-3.5-mini	ARC-Challenge	8-bit	85.8	85.5	-0.3	2.9	3.4	16	13	5
Phi-3.5-mini	ARC-Challenge	4-bit	85.8	84.2	-1.6	5.6	6.5	36	20	9
Phi-3.5-mini	MMLU	8-bit	70.8	69.7	-1.0	4.7	7.5	28	18	27
Phi-3.5-mini	MMLU	4-bit	70.8	68.8	-1.9	7.5	10.7	46	27	31
Phi-3.5-mini	CommonsenseQA	8-bit	78.8	77.8	-1.0	4.9	5.7	29	19	8
Phi-3.5-mini	CommonsenseQA	4-bit	78.8	78.4	-0.4	4.7	5.7	25	21	10

mechanism, because we measured which answers change and not why the weights behave as they do. Independent of this work, Kübler et al.^[12] argue that the accuracy thresholds used to declare quantization lossless are too loose once estimation error is taken into account, which aligns with the asymmetry we observe. A separate line of work on quantization and confidence [13] studies how the whole confidence distribution shifts, whereas our emphasis is on which answers flip and in which direction.

For someone choosing a quantization level for a small model on consumer hardware, accuracy on a benchmark is an even weaker guide than the original study warned, because accuracy can hold while a sixth of the answers change and the changes favor incorrect answers on balance. Reporting flips, the directional breakdown, and the wrong-to-wrong rate alongside accuracy would give a more complete picture of what quantization costs.

6 Limitations

Our setup differs from that of Dutta et al.^[1] in ways that prevent exact numerical replication. We evaluate zero-shot while the original MMLU results are five-shot, and we use fixed subsets in place of the full benchmarks; a version that matched the shot count and used the full benchmarks would make the comparison tighter. We test only bitsandbytes

at two bit widths, and whether the divergences also appear under GPTQ, AWQ, or other schemes remains open. We test five models below and around three billion parameters; the pattern is consistent across them, and strongest on the smallest model, but size and family are confounded in our set, and this is therefore not a controlled scaling study. Our determinism control establishes that our own measurements are not contaminated by run-to-run noise on our hardware, where the floor is zero, but it says nothing about hosted or batched serving, where the floor can be larger. Finally, we measure which answers change and characterize them by confidence, but we do not offer a mechanistic account at the level of weights or activations.

7 Conclusion

The core of Dutta et al.^[1] replicates on current small instruction-tuned models. Flips are substantial while accuracy is roughly preserved (C2), and lower-bit quantization produces more flips (C3). Two quantitative claims are weaker on this generation. Accuracy is preserved less well than the reported one to two percent (C1), and the balance between correct-to-incorrect and incorrect-to-correct flips shifts toward loss (C4). The shift remains below the level that a four-option task would produce by chance, and the recovery effect the original describes therefore persists in weaker form, while the residual tilt toward correct-to-incorrect drives the accuracy loss. The margin mechanism (C5) holds for most flips but not for all. Wrong-to-wrong transitions, which the original reports separately as AllFlips, account for a large share of behavioral change. The near-even balance the original reports for larger models is weaker on the small distilled models in common use today, and accuracy alone does not reveal the difference.

Appendix: MMLU subjects

For MMLU, we use a fixed set of eight subjects, two from each of MMLU’s four categories, to span the benchmark’s range of content instead of clustering in one area. The subjects, by dataset identifier, are `college_computer_science` and `high_school_physics` (STEM); `philosophy` and `professional_law` (humanities); `high_school_psychology` and `high_school_macroconomics` (social sciences); and `professional_medicine` and `marketing` (other). The set was fixed before any results were examined and was not revised in response to the outcomes.

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